Design of Intelligent Hive and Intelligent Bee Farm Based on Internet of Things Technology

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Abstract: Internet of things technology is to collect any object or process that needs monitoring, connection and interaction in real time through various information sensing equipment and technology. At present, in the process of bee breeding, it is the most difficult problem to understand the internal situation of the beehive under the premise of least disturbance to the bee colony. The application of Internet of things technology can solve this problem. This paper discusses the design of intelligent beehive structure, and expounds the application of iot technology in beekeeping industry from two aspects: intelligent beehive and intelligent factory. This article aims to help the beekeeping industry reduce employment, increase production efficiency and improve the quality of honey.

Key Words: Intelligent hive;Internet of things;Intelligent factory, CMOS image sensing; MCU system

1 INTRODUCTION

Internet of things (iot) technology is through a variety of information sensing equipment and technology items will be connected to the Internet (the equipment and technology including radio frequency identification system, sensors, global positioning system (GPS), infra-red sensors, laser scanner, gas sensors), according to the contract agreement to collect any need to monitor, connected, interactive objects and information, in order to realize intelligent identification, location, tracking and monitoring and management of a network technology[1]. Its purpose is to realize the connection between objects and objects, objects and people, all objects and the network, and to facilitate identification, management and control The Internet of things technology can be summarized in three aspects. First, the Internet features, that is, Internet of things that need to be connected must be able to achieve connectivity. Secondly, the recognition and communication features, that is, the "things" included in the Internet of things must have the function of automatic recognition and communication between things. Third, the intelligence features, that is, the network system should have automation, self-feedback and intelligent control characteristics. The application of iot covers all aspects of national economy and human social life. Therefore, "iot" is called the third information technology revolution after computer and Internet. The Internet of things is ubiquitous in the information age.

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With the continuous improvement of people's living standards and material quality of life, the beekeeping industry has been booming in China. However, in the process of using traditional beehives, there are always problems such as waste of honey, death of bees, difficulty in obtaining honey, and even personal harm to beekeepers under incomplete protection[3]. Therefore, the traditional beekeeping technology cannot meet the need of large-scale breeding, so it is necessary to carry out scientific planning on all aspects of beekeeping and increase the yield.Without disturbing the normal life of bees, opening the hives or disturbing the hives, it is an important problem for intelligent hives to use remote control system to manage the hives and guarantee the production of bees in a comfortable and safe environment. The Internet of things based on the Internet, realizes the connection of goods and goods, which is more and more applied in the beekeeping industry.

2 PROBLEM OF INTELLIGENT HIVE DESIGN

Whether a hive is suitable for colonies is the key to the success of beekeeping. To design a good beehive, it is necessary to solve the problem of theory, structure, volume, nestworm and material.

2.1 Theory Design

In the United States, the invention of the living box beehive developed by Lang · Strosh was a great success, which also led to the rise of the modern bee industry. However, the living cell beekeeper ignores the traditional habits of honeybees and fails to take into account the basic biological principles such as air flow, temperature and humidity in the hives, which are in urgent need of change. The design of intelligent hives should take into account the traditional habits of bees and use efficient and energy-saving insulation materials to reduce the energy consumption of the hives to maintain their temperature.

- Firstly, intelligent hives should be designed to monitor swarm behavior remotely, prevent pests in advance, and monitor population Numbers, so that beekeepers can improve their economic efficiency of bee.
- Finally, intelligent hives' temperature inside the hive and the number of bees in and out of the hive are obtained through the sensor, and the situation inside the hive is analyzed and monitored through the CMOS image sensor. Under the support of WiFi technology, intelligent bee hives will be applied to intelligent factories to reduce the employment of enterprise personnel, improve production efficiency and improve the quality of honey[4].

2.2 Structure Design

- Firstly, the traditional beehive reduces its working efficiency during operation. Therefore, the intelligent beehive must change its structure from low, short, flat and flat to upright, and automatically regulate the air flow in the beehive by means of chimney effect.
- Secondly, when the number of bees is low, the free space of traditional beehive increases the heat loss in the beehive. At this time, the temperature and humidity in the beehive need to be artificially interfered, so as to add insulation and moisturizing equipment for the beehive. Therefore, the intelligent hive must maintain the appropriate range and speed up bee population reproduction so that it reaches the maximum honey production stage.
- Finally, when honey is taken from a traditional hive, the hive's spleen is removed and sealed wax is cut off. The honey is shaken by the honey extractor, which is likely to cause death and pollution to the bees. Therefore, the intelligent hive will adopt mobile bee frames. When it is necessary to extract honey, the wall of the bee chamber will separate and use gravity to make honey flow out and enter the honey storage chamber at the bottom[5].

2.3 Volume Design

The right hive volume does not affect the bee population when it produces enough honey.

- Firstly, When using too large hives, the large number of hives cannot guarantee sufficient heat dissipation in the summer, which leads to summer decay. On the contrary, the empty interior space increases heat consumption, causing winter injuries to the bee population.
- Secondly, too small hives will hinder the reproduction and production of bees, reducing honey production.
- Some scholars suggest that the internal dimension of intelligent honeycomb is long and vertical. Its specification is 38cm by 32cm by 51cm by 32cm. It has a volume of 62,000 cm^2 and can hold a height of 45cm, a width of 28cm and a longitudinal partition.

2.4 Materia Design

- Firstly, nestworm is a major pest of honeybees, which severely interferes with their life and even causes them to fly away or lose. Nestworm depends on the wooden environment to survive. If there is less or no use of wood in the hives, nestworm cannot lay eggs in the wooden space, and the larvae have no living environment, then the nestworm problem can be effectively solved. So intelligent hives will be made from modern materials, such as Styrofoam, plastic film and aluminum foil.
- Finally, the thermal conductivity of the board outer wall used in traditional beehives is 0.1w/(m•k). It is impossible to provide insulation for beehives in winter and lack the ability to isolate external high temperatures in summer. Due to the insufficient thermal insulation performance of beehives, the bee population needs to consume a large amount of energy to keep them warm and cool every year. Therefore, the selection of foam plastics with efficient thermal insulation performance as the outer wall of intelligent beehives will effectively reduce the energy consumption of bees' thermal insulation[6].

3 IOT TECHNOLOGY APPLICATION IN INTELLIGENT HIES

During the long evolution of bee population, a unique information communication system has been developed, consisting of chemical communication, sound communication and dance communication[7]. At the same time, the change of environmental temperature directly affects the individual development, colony activity, colony reproduction, colony potential strength and other aspects of bees. The temperature and humidity of bees is the environmental factor that has the greatest impact on the bee's life activity, and the bees will actively respond to the change of environmental temperature[8]. Therefore, the detection of temperature, humidity and various behavioral characteristics in the hives can easily deal with the abnormal problems encountered in bee breeding as well as control the quality and yield of honey. The Internet of things technology is a technology to collect all kinds of necessary information such as sound, light, heat, electricity, mechanics, chemistry, biology and location in real time through all kinds of information sensing equipment and technology. Due to its real-time and interactive features, iot technology can play a significant role in intelligent beehives.At present, there are temperature monitoring, image collection and bee colony counting system which have economic and practical value to detect the physical information in hives. In the hive port with access to the bees, MSP430 ultra-low power consumption single-chip system is adopted. Combined with the dual-infrared photoelectric sensor and wireless serial port device, several systems cooperate to complete the information collection in the hive. After that, the information is sent to the server through the wireless serial port for processing, and the specific situation in the hive is identified[9]. The system design structure is as follows:

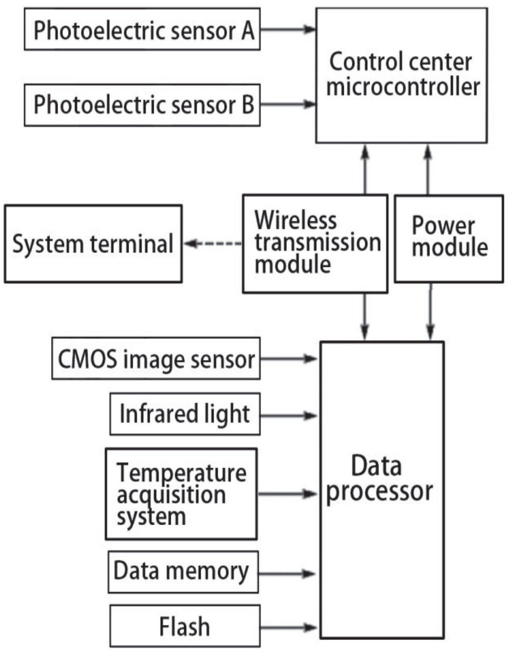

Fig 1. The architecture of intelligent hive iot system

3.1 CMOS Image Sensor System

The CMOS image sensor system is responsible for collecting bee colony image information in the hives, and it will rely on the image technology to identify the changes of colony behavior and the number of bees going in and out of the intelligent hives to obtain the living conditions of the hives.

3.2 Temperature Collection System

The temperature collection system is responsible for collecting the temperature changes in the intelligent beehive and sending them to the upper computer for analysis, so as to obtain the temperature in the beehive.

3.3 Photoelectric Sensor System

The photoelectric sensor system will pick up the number of intelligent hives that swarm in and out of the hive, and integrate the judgment of bees' behavior.

3.4 Control Center System

The control center system conducts intelligent control and managementWhen the temperature of the beehive is too high or too low, the software will start the temperature control system of the farm. When the Numbers don't match, the system automatically alerts the user to check the colony. In cases where the hives can be automatically processed, the hives are automatically processed and reported to the user. If the event cannot be handled, the user will be notified to check in time[10].

4 IOT TECHNOLOGY APPLICATION IN INTELLIGENT BEE FACTORIES

Due to the advent of the digital era and the generation of information big data, it is of great importance to realize the digitization, intelligentization and automation of the factory. If the factory can realize intelligent production and intelligent management, it can become a intelligent factory. As shown in figure 2, intelligent production and management of intelligent bee farm is realized through basic Internet of things, service Internet and information physical production system[11].

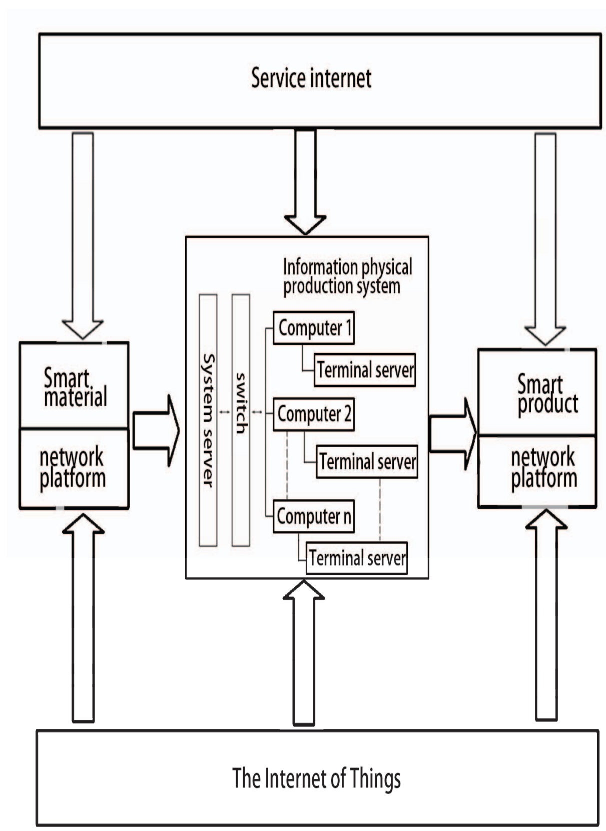

Fig 2. The architecture of intelligent factory

- Firstly,the iot system of intelligent bee farm needs to adopt control center system. As a system and tool to generate, process, transmit, store and utilize information data of bee field products, information physical production system can achieve the purpose of quality management by controlling information and data of production of peak factory products. The intelligent bee farm can build two modules: the information acquisition module and the information terminal server module , according to the actual situation of its own equipment. The information acquisition end module collects the information of factory quality management through the digital detector, and finally gathers it to the server through different information ports, for realizng the unified management of quality management information of the factory and the intellectualization of production. Information physical production system USES WiFi communication protocol and terminal interconnection. The advantage of WiFi technology is high transmission speed and long effective distance, which can save the laying of communication lines and increase the convenience of beehive. With the inherent advantage of intelligent hives' network interconnection and detection function, it is very suitable for the central regulation of a large number of hives connected by communication network in the same network, and the production and reproduction activities of hives are controlled by information physical production system. Notifying workers to go for honey when the storage chamber is nearly full can help improve honey quality and reduce energy consumption. Make changes to the environment in the factory to address the needs of buzzers, automatically deal with problems or notify staff to visit.

- The intelligent Bee Farm iot system includes basic Internet of things and service Internet which are two communication facilities. The service Internet connects suppliers and supports the information and communication integration of Planning, production, logistics, energy and management systems such as ERP and PLM. The basic Internet of things supports the interconnection between equipment, operators and products in the production process of the product, realizes the information communication of physical units such as MES, and realizes the integration with information space through CPS[12].

5 CONCLUSION

In this paper, the intelligent hives are designed to solve the problems of traditional hives' difficulty in obtaining honey, the difficulty in caring for hives, and the shortage of human hands in critical periods. The intelligent beehive has a different structure from the traditional beehive, which changes the way of breeding bees and the way of extracting honey to reduce the waste and pollution of honey. Temperature detection, CMOS image sensor and light sensor are installed in the hives to monitor the temperature and bee situation in the hives. The number of hives in and out of the hives is counted, and the temperature and humidity are automatically regulated, and the number of hives is abnormally reduced. The WiFi technology is used to connect a large number of beehives with the central information system. The intelligent Bee Farm will reduce the employment of workers, reduce the work intensity, reduce energy consumption and ensure the safety of production, so as to achieve the ultimate goal of the intelligent factory.

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